

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 80be08caf819878b

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : real_satellite_space_beacon.iq
File SHA-256 Digest : 80be08caf819878b009027f954b48b77db79af61f2bcde7c470585c99431a85b
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:29:59Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	-2.840 kHz	Spectral Centroid Offset
Estimated Baud Rate	250.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	1.46 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1746.58 kHz	Cumulative PSD 99% Energy
Estimated SNR	45.00 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	3.01 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1457	Wiener Entropy Index (0.0 – 1.0)

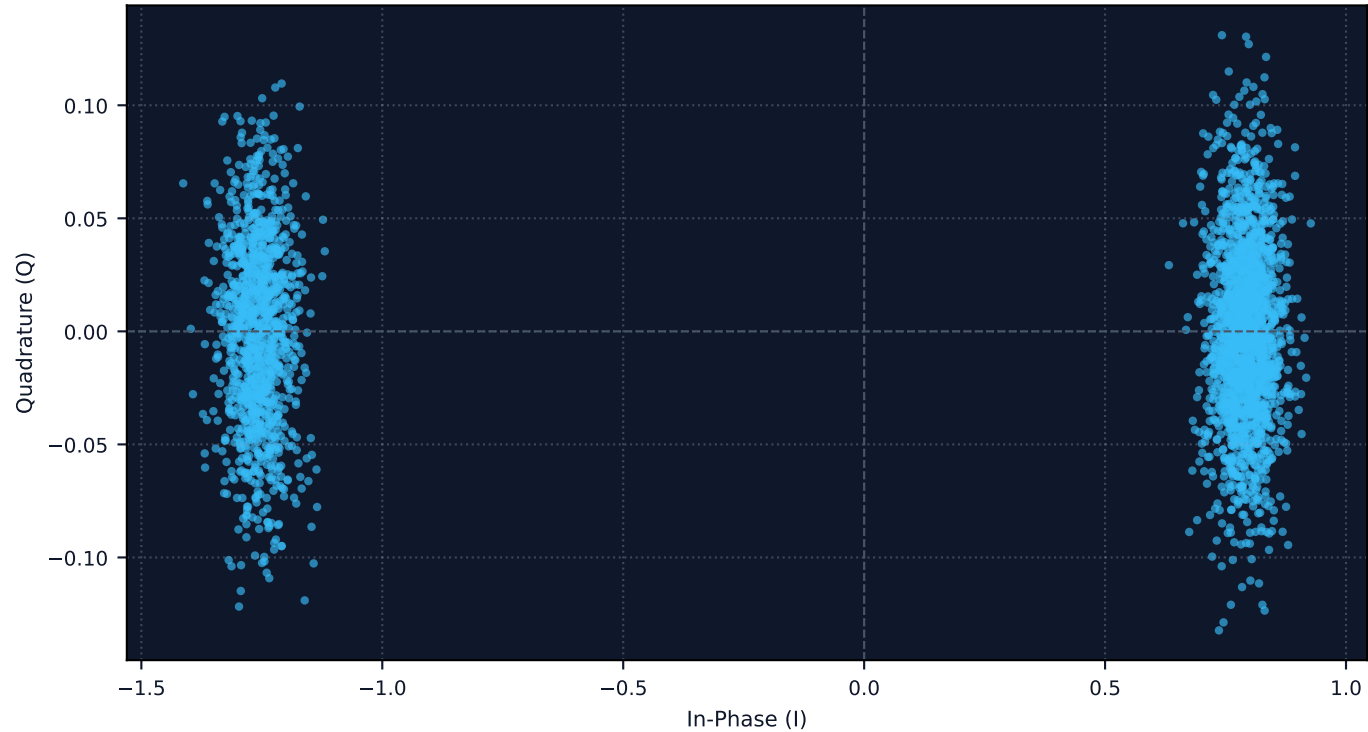
PREDICTED MODULATION : WBFM

CLASSIFIER CONFIDENCE : 93.57% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: BPSK | Recovered Bits: 8,192 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [WBFM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	0	0x0000	0	1017 B	CRC MISMATCH	PARTIAL_RECOVER
#2	285	0x0000	1	981 B	CRC MISMATCH	PARTIAL_RECOVER
#3	872	0x0000	2	908 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1157	0x0000	3	872 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1744	0x0000	4	799 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 682A88A989892A88A40848A82869E9C4

[FRAME 1 PAYLOAD]